

for all $t \in \mathbb{T}$, and $\int_0^1 (p^+(t) - p^-(t)) dt < \varepsilon$. Use this to show that if α is irrational then for any Riemann-integrable function $f : \mathbb{T} \rightarrow \mathbb{R}$,

$$\frac{1}{N} \sum_{n=0}^{N-1} f(R_\alpha^n t) \rightarrow \int f dm_{\mathbb{T}}$$

for all $t \in \mathbb{T}$.

Exercise 4.3.5. Prove Corollary 4.15

- (a) using Theorem 4.14;
- (b) using Theorem 4.10(5).

Exercise 4.3.6. ⁽⁵²⁾ Let X be a compact metric space, and let $T : X \rightarrow X$ be a continuous map. Assume that $\mu \in \mathcal{E}^T(X)$, and that for every $x \in X$ there exists a constant $C = C(x)$ such that for every $f \in C(X)$, $f \geq 0$,

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \leq C \int f d\mu.$$

Show that T is uniquely ergodic.

4.4 Measure Rigidity and Equidistribution

A natural question in number theory concerns how a sequence of real numbers is distributed when reduced modulo 1. When the terms of the sequence are generated by some dynamical process, then the expressions resemble ergodic averages, and it is natural to expect that ergodic theory will have something to offer.

4.4.1 Equidistribution on the Interval

Ergodic theorems give conditions under which all or most orbits in a dynamical system spend a proportion of time in a given set proportional to the measure of the set. In this section we consider a more abstract notion of equidistribution⁽⁵³⁾ in the specific setting of Lebesgue measure on the unit interval.

Definition 4.16. A sequence (x_n) with $x_n \in [0, 1]$ for all n is said to be *equidistributed* or *uniformly distributed* if

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f(x_k) = \int_0^1 f(x) dx \quad (4.7)$$

for any $f \in C([0, 1])$.

A more intuitive formulation (developed in Lemma 4.17) of equidistribution requires that the terms of the sequence fall in an interval with the correct frequency, just as the pointwise ergodic theorem (Theorem 2.30) says that almost every orbit under an ergodic transformation falls in a measurable set with the correct frequency.

Lemma 4.17. ⁽⁵⁴⁾ *For a sequence (x_n) of elements of $[0, 1]$, the following properties are equivalent.*

- (1) *The sequence (x_n) is equidistributed.*
- (2) *For any $k \neq 0$,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n e^{2\pi i k x_j} = 0.$$

- (3) *For any numbers a, b with $0 \leq a < b \leq 1$,*

$$\frac{1}{n} |\{j \mid 1 \leq j \leq n, x_j \in [a, b]\}| \longrightarrow (b - a)$$

as $n \rightarrow \infty$.

PROOF. (1) $\iff$ (3): Assume (1) and fix a, b with $0 \leq a < b \leq 1$. Given a sufficiently small $\varepsilon > 0$, define continuous functions that approximate the indicator function $\chi_{[a, b]}$ by

$$f^+(x) = \begin{cases} 1 & \text{if } a \leq x \leq b; \\ (x - (a - \varepsilon))/\varepsilon & \text{if } \max\{0, a - \varepsilon\} \leq x < a; \\ ((b + \varepsilon) - x)/\varepsilon & \text{if } b < x \leq \min\{b + \varepsilon, 1\}; \\ 0 & \text{for other } x, \end{cases}$$

and

$$f^-(x) = \begin{cases} 1 & \text{if } a + \varepsilon \leq x \leq b - \varepsilon; \\ (x - a)/\varepsilon & \text{if } a \leq x < a + \varepsilon; \\ (b - x)/\varepsilon & \text{if } b - \varepsilon < x \leq b; \\ 0 & \text{for other } x. \end{cases}$$

Notice that $f^-(x) \leq \chi_{[a, b]}(x) \leq f^+(x)$ for all $x \in [0, 1]$, and

$$\int_0^1 (f^+(x) - f^-(x)) \, dx \leq 2\varepsilon.$$

For small ε and $0 < a < b < 1$, these functions are illustrated in Fig. 4.2.

It follows that

$$\frac{1}{n} \sum_{j=1}^n f^-(x_j) \leq \frac{1}{n} \sum_{j=1}^n \chi_{[a, b]}(x_j) \leq \frac{1}{n} \sum_{j=1}^n f^+(x_j).$$

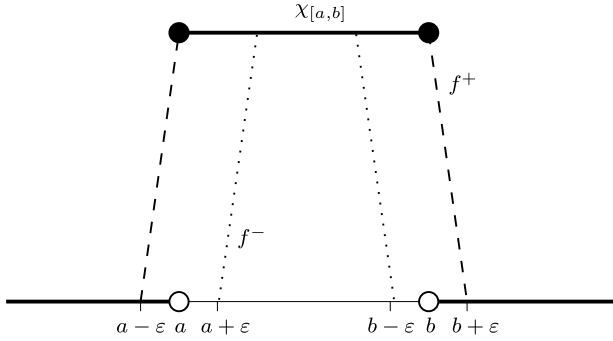


Fig. 4.2 The function $\chi_{[a,b]}$ and the approximations f^- (dots) and f^+ (dashes)

By equidistribution, this implies that

$$\begin{aligned} b - a - 2\epsilon &\leq \int_0^1 f^- \, dx \leq \liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \chi_{[a,b]}(x_j) \\ &\leq \limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \chi_{[a,b]}(x_j) \leq \int_0^1 f^+ \, dx \leq b - a + 2\epsilon. \end{aligned}$$

Thus

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \chi_{[a,b]}(x_j) = \limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \chi_{[a,b]}(x_j) = b - a$$

as required.

Conversely, if (3) holds then (1) holds since any continuous function may be approximated uniformly by a finite linear combination of indicators of intervals*.

(1) $\iff$ (2): In one direction this is clear; to see that (2) implies (1) it is enough to notice that finite trigonometric polynomials are dense in $C([0, 1])$ in the uniform metric. $\square$

Notice that equidistribution of (x_n) does not imply that (4.7) holds for measurable functions (but see Exercise 4.4.7).

Example 4.18. ⁽⁵⁵⁾ A consequence of Theorem 4.10 and Example 4.11 is that for any irrational number α , and any initial point $x \in \mathbb{T}$, the orbit $x, R_\alpha x, R_\alpha^2 x, \dots$ under the circle rotation is an equidistributed sequence. Note that this is proved in Example 4.11 by using property (2) of Lemma 4.17.

* We note that the two implications (3) $\implies$ (1) and (2) $\implies$ (1) rely on the same argument, which will be explained in detail in the proof of Corollary 4.20.

4.4.2 Equidistribution and Generic Points

Definition 4.19. If X is a compact metric space, and μ is a Borel probability measure on X , then a sequence (x_n) of elements of X is equidistributed with respect to μ if for any $f \in C(X)$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n f(x_j) = \int_X f(x) d\mu(x).$$

Equivalently, (x_n) is equidistributed if

$$\frac{1}{n} \sum_{j=1}^n \delta_{x_j} \longrightarrow \mu$$

in the weak*-topology.

For a continuous transformation $T : X \rightarrow X$ and an invariant measure μ we say that $x \in X$ is *generic* (with respect to μ and T) if the sequence of points along the orbit $(T^n x)$ is equidistributed with respect to μ . Notice that if x is generic with respect to one invariant probability measure for T , then x cannot be generic with respect to any other invariant probability measure for T . The following is an easy consequence of the ergodic theorem (Theorem 2.30).

Corollary 4.20. *Let X be a compact metric space, let $T : X \rightarrow X$ be a continuous map, and let μ be a T -invariant ergodic probability measure. Then μ -almost every point in X is generic with respect to T and μ .*

PROOF. Recall that $C(X)$ is a separable metric space with respect to the uniform norm

$$\|f\|_\infty = \sup\{|f(x)| \mid x \in X\}$$

by Lemma B.8. Let $(f_n)_{n \geq 1}$ be a dense sequence in $C(X)$. By applying Theorem 2.30 to each of these functions we obtain one set $X' \subseteq X$ of full measure with the property that

$$\frac{1}{N} \sum_{n=0}^{N-1} f_i(T^n x) \longrightarrow \int_X f_i d\mu$$

for all $i \geq 1$ and $x \in X'$. Now let $f \in C(X)$ be any function and fix $\varepsilon > 0$. By the uniform density of the sequence, we may find an $i \in \mathbb{N}$ for which

$$|f(x) - f_i(x)| < \varepsilon$$

for all $x \in X$. Then

$$\int f \, d\mu - 2\varepsilon \leq \liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \leq \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \leq \int f \, d\mu + 2\varepsilon,$$

showing convergence of the ergodic averages for f at any $x \in X'$. The limit must be $\int f \, d\mu$ since $|\int f \, d\mu - \int f_i \, d\mu| \leq \varepsilon$, so x is a generic point. $\square$

4.4.3 Equidistribution for Irrational Polynomials

Example 4.18 may be thought of as a statement in number theory: for an irrational α , the values of the polynomial $p(n) = x + \alpha n$, when reduced modulo 1, form an equidistributed sequence for any value of x . Weyl [381] generalized this to more general polynomials, and Furstenberg [98] found that this result could also be understood using ergodic theory. We recall the statement of Weyl's polynomial equidistribution Theorem (Theorem 1.4 on p. 4): Let $p(n) = a_k n^k + \cdots + a_0$ be a real polynomial with at least one coefficient among $a_1, \dots, a_k$ irrational. Then the sequence $(p(n))$ is equidistributed modulo 1.

As indicated in Example 4.18, the unique ergodicity of irrational circle rotations proves Theorem 1.4 for $k = 1$. More generally, Theorem 4.10 shows that the orbits of any transformation of the circle for which the Lebesgue measure is the unique invariant measure are equidistributed. In order to apply this to the case of polynomials, we turn to a structural result of Furstenberg [99] that allows more complicated transformations to be built up from simpler ones while preserving a dynamical property (in Chap. 7 a similar approach will be used for another application of ergodic theory).

Notice that by Theorem 4.10, orbits of a uniquely ergodic transformation are equidistributed with respect to the unique invariant measure.

Theorem 4.21 (Furstenberg). *Let $T : X \rightarrow X$ be a uniquely ergodic homeomorphism of a compact metric space with unique invariant measure μ . Let G be a compact group^{*} with Haar measure m_G , and let $c : X \rightarrow G$ be a continuous map. Define the skew-product map S on $Y = X \times G$ by*

$$S(x, g) = (T(x), c(x)g).$$

If S is ergodic with respect to $\mu \times m_G$, then it is uniquely ergodic.

PROOF. To see that S preserves $\mu \times m_G$, let $f \in C(Y)$. Then, by Fubini's theorem,

^{*} The reader may replace G by a torus $\mathbb{T}^k$ with group operation written additively, together with Lebesgue measure $m_{\mathbb{T}^k}$. Notice that in any case the Haar measure is invariant under multiplication on the right or the left since G is compact (see Sect. C.2).

$$\begin{aligned}
\int_Y f \circ S \, d(\mu \times m_G) &= \int_X \int_G f(Tx, c(x)g) \, dm_G(g) \, d\mu(x) \\
&= \int_X \int_G f(Tx, g) \, dm_G(g) \, d\mu(x) \\
&= \int_X \int_G f(x, g) \, dm_G(g) \, d\mu(x) = \int_Y f \, d(\mu \times m_G).
\end{aligned}$$

Assume that S is ergodic. Let

$$E = \{(x, g) \mid (x, g) \text{ is generic w.r.t. } \mu \times m_G\}.$$

By Corollary 4.20, $\mu \times m_G(E) = 1$. We claim that E is invariant under the map $(x, g) \mapsto (x, gh)$. To see this, notice that $(x, g) \in E$ means that

$$\frac{1}{N} \sum_{n=0}^{N-1} f(S^n(x, g)) \longrightarrow \int f \, d(\mu \times m_G)$$

for all $f \in C(X \times G)$. Writing $f_h(\cdot, g) = f(\cdot, gh)$, it follows that

$$\begin{aligned}
\frac{1}{N} \sum_{n=0}^{N-1} f(S^n(x, gh)) &= \frac{1}{N} \sum_{n=0}^{N-1} f_h(S^n(x, g)) \\
&\longrightarrow \int f_h \, d(\mu \times m_G) = \int f \, d(\mu \times m_G)
\end{aligned}$$

since m_G is invariant under multiplication on the right, so $(x, gh) \in E$ also. It follows that $E = E_1 \times G$ for some set $E_1 \subseteq X$, $\mu(E_1) = 1$. Now assume that ν is an S -invariant ergodic measure on Y . Write $\pi : Y \rightarrow X$ for the projection $\pi(x, g) = x$. Then $\pi_*\nu$ is a T -invariant measure, so by unique ergodicity $\pi_*\nu = \mu$. In particular, $\nu(E) = \nu(E_1 \times G) = \mu(E_1) = 1$. By Corollary 4.20, ν -almost every point is generic with respect to ν . Thus there must be a point $(x, g) \in E$ generic with respect to ν . By definition of E , it follows that $\nu = \mu \times m_G$. $\square$

Corollary 4.22. *Let α be an irrational number. Then the map $S : \mathbb{T}^k \rightarrow \mathbb{T}^k$ defined by*

$$S : \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix} \longmapsto \begin{pmatrix} x_1 + \alpha \\ x_2 + x_1 \\ \vdots \\ x_k + x_{k-1} \end{pmatrix}$$

is uniquely ergodic.

PROOF. Notice that the transformation S is built up from the irrational circle map by taking $(k-1)$ skew-product extensions as in Theorem 4.21. By Theorem 4.21, it is sufficient to prove that S is ergodic with respect to

Lebesgue measure on $\mathbb{T}^k$. Let $f \in L^2(\mathbb{T}^k)$ be an S -invariant function, and write

$$f(\mathbf{x}) = \sum_{\mathbf{n} \in \mathbb{Z}^k} c_{\mathbf{n}} e^{2\pi i \mathbf{n} \cdot \mathbf{x}}$$

for the Fourier expansion of f . Then, since $f(\mathbf{x}) = f(S\mathbf{x})$, we have

$$\sum_{\mathbf{n} \in \mathbb{Z}^k} c_{\mathbf{n}} e^{2\pi i \mathbf{n} \cdot S\mathbf{x}} = \sum_{\mathbf{n} \in \mathbb{Z}^k} c_{\mathbf{n}} e^{2\pi i n_1 \alpha} e^{2\pi i S' \mathbf{n} \cdot \mathbf{x}}$$

where

$$S' : \begin{pmatrix} n_1 \\ n_2 \\ \vdots \\ n_{k-1} \\ n_k \end{pmatrix} \mapsto \begin{pmatrix} n_1 + n_2 \\ n_2 + n_3 \\ \vdots \\ n_{k-1} + n_k \\ n_k \end{pmatrix}$$

is an automorphism of $\mathbb{Z}^k$. By the uniqueness of Fourier coefficients,

$$c_{S'\mathbf{n}} = e^{2\pi i \alpha n_1} c_{\mathbf{n}}, \quad (4.8)$$

and in particular $|c_{S'\mathbf{n}}| = |c_{\mathbf{n}}|$ for all $\mathbf{n}$. Thus for each $\mathbf{n} \in \mathbb{Z}^k$ we either have $\mathbf{n}, S'\mathbf{n}, (S')^2\mathbf{n}, \dots$ all distinct (in which case $c_{\mathbf{n}} = 0$ since $\sum_{\mathbf{n}} |c_{\mathbf{n}}|^2 < \infty$) or $(S')^p \mathbf{n} = (S')^q \mathbf{n}$ for some $p > q$, so $n_2 = n_3 = \dots = n_k = 0$ (by downward induction on k , for example). Now for $\mathbf{n} = (n_1, 0, \dots, 0)$, (4.8) simplifies to $c_{\mathbf{n}} = e^{2\pi i n_1 \alpha} c_{\mathbf{n}}$, so $n_1 = 0$ or $c_{\mathbf{n}} = 0$. We deduce that f is constant, so S is ergodic. $\square$

PROOF OF THEOREM 1.4. Assume that Theorem 1.4 holds for all polynomials of degree strictly less than k . If a_k is rational, then $qa_k \in \mathbb{Z}$ for some integer q . Then the quantities $p(qn+j)$ modulo 1 for varying n and fixed $j = 0, \dots, q-1$, coincide with the values of polynomials of degree strictly less than k satisfying the hypothesis of the theorem. It follows that the values of each of those polynomials are equidistributed, so the values of the original polynomial are equidistributed modulo 1 by induction. Therefore, we may assume without loss of generality that the leading coefficient a_k is irrational.

A convenient description of the transformation S in Corollary 4.22 comes from viewing $\mathbb{T}^k$ as $\{\alpha\} \times \mathbb{T}^k$ with a map defined by

$$\begin{pmatrix} 1 & & & & \\ 1 & 1 & & & \\ & 1 & 1 & & \\ & & \ddots & \ddots & \\ & & & 1 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix} = \begin{pmatrix} \alpha \\ x_1 + \alpha \\ x_2 + x_1 \\ \vdots \\ x_k + x_{k-1} \end{pmatrix}.$$

Iterating this map gives

$$\begin{aligned}
\begin{pmatrix} 1 & & & \\ 1 & 1 & & \\ & 1 & 1 & \\ & & \ddots & \\ & & & 1 & 1 \end{pmatrix}^n \begin{pmatrix} \alpha \\ x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix} &= \begin{pmatrix} 1 & & & \\ n & 1 & & \\ \binom{n}{2} & n & 1 & \\ \vdots & & \ddots & \ddots \\ \binom{n}{k} & \dots & n & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix} \\
&= \begin{pmatrix} \alpha \\ n\alpha + x_1 \\ \binom{n}{2}\alpha + nx_1 + x_2 \\ \vdots \\ \binom{n}{k}\alpha + \binom{n}{k-1}x_1 + \dots + nx_{k-1} + x_k \end{pmatrix}.
\end{aligned}$$

Now define $\alpha = k!a_k$, and choose points $x_1, \dots, x_k$ so that

$$p(n) = \binom{n}{k}\alpha + \binom{n}{k-1}x_1 + \dots + nx_{k-1} + x_k.$$

Then by Corollary 4.22, the orbits of this map are equidistributed on $\mathbb{T}^k$, so the same holds for its last component, which coincides with the sequence of values of $p(n)$ reduced modulo 1 in $\mathbb{T}$. $\square$

An alternative approach in the quadratic case will be described in Exercise 7.4.2.

Exercises for Sect. 4.4

Exercise 4.4.1. Consider the circle-doubling map $T_2 : x \mapsto 2x \pmod{1}$ on $\mathbb{T}$ with Lebesgue measure $m_{\mathbb{T}}$.

- (a) Construct a point that is generic for $m_{\mathbb{T}}$.
- (b) Construct a point that is generic for a T_2 -invariant ergodic measure other than $m_{\mathbb{T}}$.
- (c) Construct a point that is generic for a non-ergodic T_2 -invariant measure.
- (d) Construct a point that is not generic for any T_2 -invariant measure.

Exercise 4.4.2. Extend Lemma 4.17 to show that (4.7) holds for Riemann-integrable functions (cf. Exercise 4.3.4). Could it hold for Lebesgue-integrable functions?

Exercise 4.4.3. Use Exercise 4.3.4 to show that the fractional parts of the sequence $(n\alpha)$ are uniformly distributed in $[0, 1]$. That is,

$$\frac{|\{n \mid 0 \leq n < N, n\alpha - \lfloor n\alpha \rfloor \in [a, b)\}|}{N} \rightarrow (b - a)$$

as $N \rightarrow \infty$, for any $0 \leq a < b \leq 1$.

Exercise 4.4.4. Carry out the procedure used in the proof of Theorem 1.4 to prove that the sequence (x_n) defined by $x_n = \begin{pmatrix} \alpha_1 n \\ \alpha_2 n^2 \end{pmatrix}$ is equidistributed in $\mathbb{T}^2$ if and only if $\alpha_1, \alpha_2 \notin \mathbb{Q}$.

Exercise 4.4.5. A number α is called a *Liouville number* if there is an infinite sequence $(\frac{p_n}{q_n})_{n \geq 1}$ of rationals with the property that

$$\left| \frac{p_n}{q_n} - \alpha \right| < \frac{1}{q_n^n}$$

for all $n \geq 1$. Notice that Exercise 3.3.3 shows that algebraic numbers are not Liouville numbers.

(a) Assuming that α is not a Liouville number, prove the following error rate in the equidistribution of the sequence $(x + n\alpha)_{n \geq 1}$ modulo 1:

$$\left| \frac{1}{N} \sum_{n=0}^{N-1} f(x + n\alpha) - \int_0^1 f(x) dx \right| \leq S(\alpha, f) \frac{1}{N},$$

for $f \in C^\infty(\mathbb{T})$ and some constant $S(\alpha, f)$ depending on α and f .

(b) Formulate and prove a generalization to rotations of $\mathbb{T}^d$.

Exercise 4.4.6. Use the ideas from Exercise 2.8.4 to prove a mean ergodic theorem along the squares: for a measure-preserving system $(X, \mathcal{B}, \mu, T)$ and $f \in L^2_\mu$, show that

$$\frac{1}{N} \sum_{n=0}^{N-1} U_T^{n^2} f$$

converges in L^2_μ . Under the assumption that T is totally ergodic (see Exercise 2.5.6), show that the limit is $\int f d\mu$.

Exercise 4.4.7. Let X be a compact metric space, and assume that $\nu_n \rightarrow \mu$ in the weak*-topology on $\mathcal{M}(X)$. Show that for a Borel set B with $\mu(\partial B) = 0$,

$$\lim_{n \rightarrow \infty} \nu_n(B) = \mu(B).$$

Notes to Chap. 4

⁽⁴⁸⁾(Page 98) The fact that $\mathcal{M}^T(X)$ is non-empty may also be seen as a result of various fixed-point theorems that generalize the Brouwer fixed point theorem to an infinite-dimensional setting; the argument used in Sect. 4.1 is attractive because it is elementary and is connected directly to the dynamics.

⁽⁴⁹⁾(Page 103) A convenient source for the Choquet representation theorem is the updated lecture notes by Phelps [283]; the original papers are those of Choquet [55, 56].

⁽⁵⁰⁾(Page 103) Notice that the space of invariant measures for a given continuous map is a topological attribute rather than a measurable one: measurably isomorphic systems may

have entirely unrelated spaces of invariant measures. In particular, the Jewett–Krieger theorem shows that any ergodic measure-preserving system $(X, \mathcal{B}, \mu, T)$ on a Lebesgue space is measurably isomorphic to a minimal, uniquely ergodic homeomorphism on a compact metric space (a continuous map on a compact metric space is called *minimal* if every point has a dense orbit; see Exercise 4.2.1). This deep result was found by Jewett [166] for weakly-mixing transformations, and was extended to ergodic systems by Krieger [213] using his proof of the existence of generators [212]. Thus having a model (up to measurable isomorphism) as a uniquely ergodic map on a compact metric space carries no information about a given measurable dynamical system. Among the many extensions and modifications of this important result, Bellow and Furstenberg [22], Hansel and Raoult [140] and Denker [69] gave different proofs; Jakobs [164] and Denker and Eberlein [70] extended the result to flows; Lind and Thouvenot [231] showed that any finite entropy ergodic transformation is isomorphic to a homeomorphism of the torus $\mathbb{T}^2$ preserving Lebesgue measure; Lehrer [222] showed that the homeomorphism can always be chosen to be topologically mixing (a homeomorphism $S : Y \rightarrow Y$ of a compact metric space is topologically mixing if for any open sets $U, V \subseteq Y$, there is an $N = N(U, V)$ with $U \cap S^n V \neq \emptyset$ for $n \geq N$); Weiss [379] extended to certain group actions and to diagrams of measure-preserving systems; Rosenthal [317] removed the assumption of invertibility. In a different direction, Downarowicz [74] has shown that every possible Choquet simplex arises as the space of invariant measures of a map even in a highly restricted class of continuous maps.

⁽⁵¹⁾(Page 104) Birkhoff’s recurrence theorem may be thought of as a topological analog of Poincaré recurrence (Theorem 2.11), with the essential hypothesis of finite measure replaced by compactness. Furstenberg and Weiss [109] showed that there is also a topological analog of the ergodic multiple recurrence theorem (Theorem 7.4): if (X, T) is minimal and $U \subseteq X$ is open and non-empty, then for any $k > 1$ there is some $n \geq 1$ with

$$U \cap T^n U \cap \dots \cap T^{(k-1)n} U \neq \emptyset.$$

⁽⁵²⁾(Page 110) This characterization is due to Pjateckiĭ–Šapiro [285], who showed it as a property characterizing normality for orbits under the map $x \mapsto ax \pmod{1}$.

⁽⁵³⁾(Page 110) The theory of equidistribution from the viewpoint of number theory is a large and sophisticated one. Extensive overviews of this theory in three different decades may be found in the monographs of Kuipers and Niederreiter [215], Hlawka [154], and Drmota and Tichy [75].

⁽⁵⁴⁾(Page 111) The formulation in (2) is the Weyl criterion for equidistribution; it appears in his paper [381]. Weyl really established the principle that equidistribution can be shown using a sufficiently rich set of test functions; in particular on a compact group it is sufficient to use an appropriate orthonormal basis of L^2 . Thus a more general formulation of the Weyl criterion is as follows. Let G be a compact metrizable group and let $G^\#$ denote the set of conjugacy classes in G . Then a sequence (g_n) of elements of $G^\#$ is equidistributed with respect to Haar measure if and only if

$$\sum_{j=1}^n \operatorname{tr}(\pi(g_j)) = o(n)$$

as $n \rightarrow \infty$, for any non-trivial irreducible unitary representation $\pi : G \rightarrow \operatorname{GL}_k(\mathbb{C})$. For more about equidistribution in the number-theoretic context, see the monograph of Iwaniec and Kowalski [162, Ch. 21].

⁽⁵⁵⁾(Page 112) This equidistribution result was proved independently by several people, including Weyl [380], Bohl [39] and Sierpiński [344].

